# Tanzim Hossain Romel

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**Research interests:** AI for software engineering and reliability, trustworthy AI, and long-horizon coding agents.

## Education

- **University of Alberta** *Sep 2026 - Present*
  *M.Sc. in Computing Science* Edmonton, Canada
  - Thesis-based M.Sc. supervised by Dr. Zhou Yang.

- **Bangladesh University of Engineering and Technology (BUET)** *Mar 2018 - May 2023*
  *B.Sc. in Computer Science and Engineering* Dhaka, Bangladesh

## Research Experience

- **University of Alberta** *Sep 2026 - Dec 2026*
  *Graduate Research Assistantship Fellowship (GRAF)* Edmonton, Canada
  - Research reliable LLM coding agents, with a focus on behavior steering and code maintainability over long tasks, under Dr. Zhou Yang's supervision.

- **University of Illinois Urbana-Champaign (UIUC)** *Jun 2026 - Aug 2026*
  *Remote Research Intern, UIUC++ SRSE 2026, Professor Tianyin Xu's lab*
  - Curated real-world postmortems into reproducible SREGym benchmark problems.
  - Built Kubernetes scenarios with fault injection, mitigation oracles, and checks that reject shortcuts.

## Selected Research Papers

- **Choice as an Attack: Auditing Aligned Backdoor Attacks in LLM Agents** *Aug 2025 - Present*
  *Submitted to ACL Rolling Review (ARR), 2026*
  - Implemented **SHIFT**'s matched trigger/no-trigger audits with clean-model controls, permutation tests, and conditional-logit models that account for option quality.
  - Added product-order shuffles, reversible brand aliases, and option-level logging of price, rating, display position, and target labels.
  - Built evaluation configurations for Qwen, Ministral, and Llama across controlled shopping and two-tool routing tasks.

- **An Empirical Study on Remote Code Execution in Machine Learning Model Hosting Ecosystems** *Jun 2025 - Oct 2025*
  *Preprint: arXiv:2601.14163*
  - Analyzed about 45,000 ML repositories across five hosting platforms for unsafe deserialization, code injection, and malware risks.
  - Analyzed 600+ developer discussions to classify misconceptions about remote code execution and compared platform defenses.

## Teaching Experience

- **University of Alberta** *Sep 2026 - Present*
  *Graduate Teaching Assistant (GTA)* Edmonton, Canada
  - Teach introductory computing labs for CMPUT 174, sections D15 and Online Lab 804 (Fall 2026).
  - Conduct lab check-ins, assess student code demonstrations, answer questions, and grade lab work.

## Professional Experience

- **IQVIA** *Jun 2023 - Jun 2026*
  *Software Development Engineer* Dhaka, Bangladesh
  - Built and maintained KPI Library, IQVIA's healthcare analytics reporting layer, using C#/.NET, EF Core, and AWS.
  - Optimized selected analytics paths, reducing complex query time by 60% and API latency by 40%.
  - Built LangGraph/LangChain workflows for dashboard setup and analysis, reducing setup effort by 85% in the targeted workflow.

## Selected Research Software

- **ctxhelm + HelmBench** *May 2026 - Present*
  - Built a read-only context compiler for coding agents using lexical and symbol search, dependency expansion, and related-test mapping.
  - Built evaluation infrastructure with trace schemas, evidence bundles, and quality checks that separate context selection from patch quality without exposing source or prompts.
- **SchemaAgent: Relational Schema and ER Diagram Generation** *Jul 2025 - Present*
  - Extended SchemaAgent with Dr. Sukarna Barua (BUET), using LangGraph and Z3 to generate and validate relational schemas from natural-language requirements.

## Selected Open Source Contributions

2025 - 2026

- **RefactoringMiner**: merged MCP tooling, WebDiff support, merge-parent-aware commit diffs, and assertion migration matching.
- **EF Core**: merged changes to runtime migration creation/application, `ON DELETE SET DEFAULT` support, and nullable complex-property reload handling.
- **TypeScript's Go port**: merged a declaration-emit fix that preserves imported type aliases for inferred exports across module boundaries.

## Technical Skills

**Languages and systems:** Python, C#/.NET, Rust, TypeScript, Go, C++, PostgreSQL, AWS, Docker, Kubernetes.

**Research and agent tooling:** PyTorch, vLLM, LangGraph, MCP, CodeQL, Semgrep, Bandit, YARA, Z3.

## Selected Honors

- **IQVIA Impact Program Silver Award** for essential feature delivery.
- **2nd among 120 students**, BUET Bangla handwritten-digit recognition contest, with 95.9% accuracy.
- **Dean's List Award**, BUET Level 2.